

What the data already says

Peterbilt Atlantic. Patterns found in the existing records, without being told what to look for.

Draft v0.1 · 24 August 2026 · Not approved. Prepared for Tobias.

On 21 August the ask was to find *patterns we cannot define in advance*. So nothing on this page was requested by anybody, and none of it began with a metric somebody had already thought of.

How this was produced, in short

Rather than deciding in advance what to look for, **eight separate automated analyses were run over the same records at once**, each given a different lens: complaint language, contact rhythm, parts vocabulary, customers going quiet, voicemail, the calendar, joining phone to email, and one given no instruction beyond “what is strange here?”

They were kept apart on purpose. None could see another’s work. That way, agreement between them means something.

Between them they produced 75 findings. The ones that appear on this page are mainly the ones *several analyses reached independently*, from different data and by different arithmetic. Those are the ones worth acting on. A few were checked afterwards, found to be wrong, and removed, and this page says where that happened.

All of that came from reading 0.26% of the email. The same method over the complete records would find considerably more, and would settle several questions this page has to leave open.

How to read this page. Please read this first

🕒 COUNTED: ALL OF IT

We counted every single record. Nothing was estimated, sampled or guessed. All 2,278,720 emails, all 24,560 scored calls, all 131,038 customer records.

You can quote these numbers to anyone. More data would not change them.

📖 READ: A SAMPLE ONLY

We read the actual words in a small slice. To understand *what people say*, somebody has to read the text, and only **0.26% of the emails** (10,320 of 2,278,720) and 14.5% of call transcripts were read.

These findings are real but incomplete. They show the shape of the answer. **The full corpus would sharpen them, and is what we are asking for.**

Why the split exists at all. Counting is cheap: the database can count two million emails in seconds without sending them anywhere. Reading is expensive: the text has to be pulled out and worked through. So we counted everything, and read a carefully chosen slice. Every number below says which one it is.

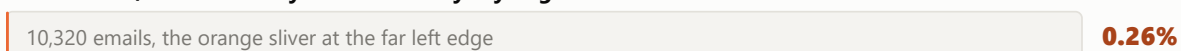
How much was counted, and how much was read

Each bar is the same 2,278,720 emails.

Counted, used for every “how many” figure



Read as text, used for every “what did they say” figure



The orange sliver is drawn to scale. **Everything in Section 1 of this page was found inside that sliver, or by counting.** That is the argument for access to the rest.

1. The dealership loses the follow-up

Found independently by **4 of the 8** analyses. The strongest result on this page.

59.7%

of the voicemails we can check never get a callback.

368 of 616, Peterbilt only, robocalls removed. The ones that *were* returned took a median of **7 days**. The ones that were not have now been waiting a median of **109 days**, and counting.

Important limit, found by checking one real number against the screen: this covers only half the voicemails. See the box directly below.

🕒 COUNTED: ALL OF IT

Half of the voicemails cannot be checked at all, and nobody knew.

Peterbilt has **1,232 voicemails** in the system. The report that works out whether somebody rang back can only see **669** of them. The other **563, which is 45.7%, are invisible to it.**

The reason is mechanical. Those 563 arrived without a call reference attached, so there is nothing to match them against the outgoing call records. They are not counted as returned. They are not counted as ignored. They simply fall out of the report.

So the 59.7% above describes the half we can see. For the other half we genuinely do not know, and any claim about them would be invented. This was found by taking one real customer number and checking it against the screen, which is worth more than any amount of further analysis.

"A real message", about one in five voicemails in the raw count is a few seconds of dead air, where somebody rang off without speaking. Those were removed before calculating

this. The 59.7% is of genuine messages only, so it is the conservative number, not the dramatic one.

7 of 9

"Truck down or driver stranded" voicemails never returned, *of the ones the report can see*

3,694

Genuine repeat callers, robocalls excluded, who never got a callback

53

Inbound calls from one single Peterbilt number, with no callback ever logged

326

Strongly negative calls, none of which can be tied to a tracked customer record

"I'd like to get this truck back on the road as soon as possible. I've already lost \$30,000 with it sitting there."

Ryan Walsh, voicemail left 25 June 2026, urgency scored 0.95 out of 1. Verbatim from the recording. **We cannot say whether this particular message was returned**, because it does not appear in the callback records either way. It is quoted to show what an urgent message sounds like, not as an example of one that was ignored.

Two things make this worse than it first looks

1. It is invisible in the system the business actually uses. 326 Peterbilt calls are scored strongly negative. **Not one of them** can be traced to a tracked customer record, because phone and email do not join at all (Section 4). A customer can ring up furious and leave no mark at all on the file anybody looks at.

🕒 **COUNTED: ALL OF IT**

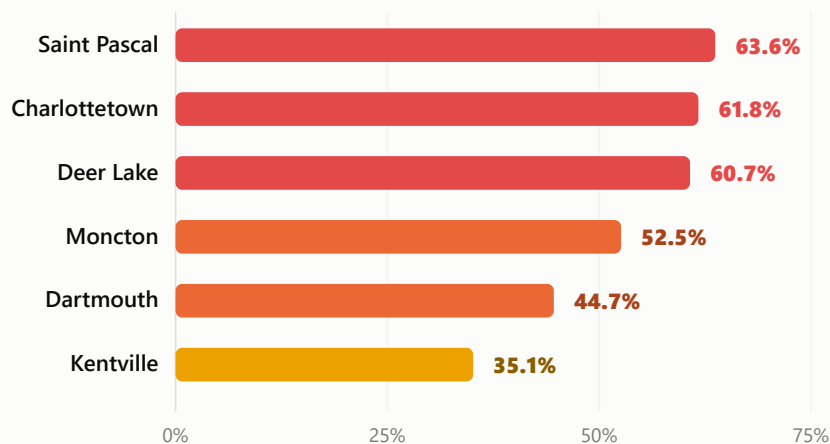
2. Listening for angry customers would not find these people. Callers who repeatedly get no callback **do not sound angrier**, measured, their tone is slightly *calmer* than average. They have given up rather than escalated.

Why this matters for what we build: the obvious product, “flag the upset customers”, would systematically miss exactly the customers being failed worst. The right measure is *whether anyone called them back*, not how they sounded.

🕒 COUNTED: ALL OF IT

It varies by store, but less dramatically than it first looked

Voicemails with no callback recorded, by site. Only the voicemails the report can see, so read this alongside the blind spot above.



🕒 COUNTED: ALL OF IT Sites with at least 20 voicemails. Kentville is the best at 35.1% and Saint Pascal the worst at 63.6%, **so there is a real spread of roughly 1.8 times, not the seven times an earlier draft of this page claimed.** A further 1,058 voicemails carry no site at all, which is the largest group of the lot.

Calls drift towards voicemail as the day goes on

Share of incoming calls that end in voicemail, by local hour. Peterbilt is open 07:00 to 17:00.



200,000 calls. The rise is real but gentle: about 1.7 times across the day.

🕒 COUNTED: ALL OF IT

Opening hours are not a guess: `twin.tenant_hours` records Peterbilt as 07:00 to 17:00, and migration 016 notes those were derived from the observed call volume by hour, not from anyone's recollection. Times converted from UTC to Atlantic. **An earlier draft of this page overstated this as a sharp cliff in the last two hours. It is a gradual drift, and the sharpest single hour is the closing hour itself.**

What this becomes: a callback-owed list, ordered by how long the customer has been waiting and whether they said a truck was down.

This needs no new data and no CDK access. It can be built now.

2. Customers announce that they are leaving, months in advance

Found independently by **2 of the 8** analyses, using different arithmetic and arriving at the same answer.

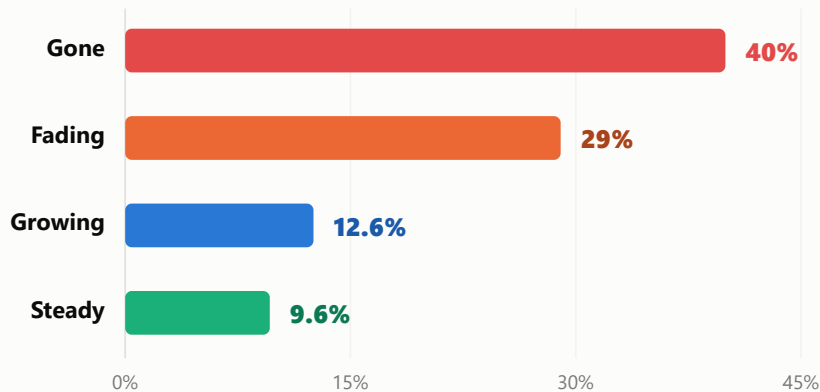
The question was: when a customer stops dealing with the business, is there anything in the record *beforehand* that says it is coming? There is, and it is silence.

Read this carefully, because it is the easy thing to get wrong. We did not measure "quiet customers are quiet", that would be circular and meaningless. We walked each

account backwards from **its own final contact**, whenever that was, and asked what the months before it looked like. A healthy account measured the same way is the comparison.

Went three straight silent months before their final-ever contact

Each account measured against its own history, not against today's date.



🕒 COUNTED: ALL OF IT

The gap between the red bars and the blue-green ones is the whole finding: **this happens four times more often to customers who are about to leave**. The median departing account had *zero* contact in every one of the seven months before its last.

The second analysis, which never saw the first, measured it a different way, by how long the gaps between contacts were rather than by counting silent months. It found departing accounts were **3.7 times more likely** (29% against 8%) to show one unusually long pause first.

Two methods, two analyses, no contact between them, same answer. That is why this one is trustworthy.

What this becomes: the client feed. "Fleet Services has been silent 3 months. It was monthly for two years."

Also buildable now.

3. Customers use words the system does not listen for

8.2% vs 0%

“Derate” appears in 8.2% of urgent calls and in *none at all* of the calm ones.

892 urgent calls, 1,211 calm ones, zero uses of the word among the calm. By comparison the word “emergency” is only 7 times more common in urgent calls. Derate is the cleaner signal of the two.

 **READ: A SAMPLE ONLY** from 3,627 transcripts. **The full set would confirm it and find the others like it.**

What “derate” means. A modern truck detects a fault, usually in its emissions system, and electronically limits its own power to protect itself. The truck is still running but crawls, sometimes at 5 mph. It is not broken down, so it does not sound like an emergency in ordinary language. To the driver it is one: a loaded truck that cannot make its delivery. **No off-the-shelf system would ever weight this word. It is specific to this trade.**

8.2%

Of sampled calls use emission-system vocabulary, and they run **46% more urgent**

 **COUNTED**

97s → 209s

Median call length, calm (n=966) versus very negative (n=326). **Anger has a length.**

 **COUNTED**

8–15

Calls out of **24,560** where the existing tags say the customer was upset

 **COUNTED**

149

Transcripts where the customer mentions **Kenworth**, the sister brand

 **SAMPLE**

Customers are shopping the sister brand out loud, on recorded calls. Kenworth is PACCAR's other marque, the same parent company as Peterbilt. It is named more than any other brand in the transcripts we read.

Read alongside Section 1, the mechanism is not subtle: when nobody gets back to them, customers say so, and they say where they went instead. **In the sample we could read, this appeared 149 times. We cannot say how often it happens across all 25,100 calls, because we have only read 14.5% of them.**

4. Why we cannot yet answer the money question

This is the case for CDK, in one measurement.

Can a customer be followed from email into phone calls?

Matching 21,453 verified email accounts against every phone record.



In practice: zero customers can be followed across both.

🔍 **COUNTED: ALL OF IT**

Two records of the same customers, with nothing linking them.

Why it fails. An email record knows an address. A phone record knows a number. Nothing in either says they are the same company. **An invoice knows both**, it carries the customer's identity, their email, and what they actually spent. That is why CDK is not simply a third source of data. It is the key the other two are missing.

The consequence today: every finding on this page is about *contact*. None is about *money*. We can say a voicemail was never returned; we cannot say what it cost. With invoices joined in:

- Unreturned voicemails become **revenue at risk**, by customer and by site.
- Those Kenworth mentions become **measurable leakage**, did that customer's spend actually fall afterwards?
- A quiet account gets an **amount** attached, which is what turns an alert into a priority.
- The gap between the best and worst branch becomes **a cost**, and the business case writes itself.

5. What the product already handles, and the small things left

We went looking for a problem here and mostly found it already solved. That is worth saying plainly.

This was already found, and already fixed, before we looked.

The screen does not claim 131,038 customers. It says **6,984**, and it gets there through filters somebody built deliberately.

How 131,038 records become 6,984 customers

Each step is a filter already in the product today.

131,038 records in total		131,038
	only companies, not individual people	21,457
	at least 5 pieces of mail, so a stray reply is not an account	7,983
	automated senders removed by hand (25 of them)	7,958
	enough history to say which way they are going	6,984

 **COUNTED: ALL OF IT** Every step verified against the live database and the code that draws the page.

How the automated senders were identified, which is the clever part. The obvious approach is a list of banned names, and it quietly deletes real customers: `dexter.ca` sounds like a parts supplier and is actually Dexter Construction, a real fleet. So instead the rule uses **whether the dealership ever writes back**. Measured: `cdkmessaging.com` 849 messages and **0 replies**; `credifax.com` 1,828 messages and 43 replies, 2%; against `dexter.ca` 541 messages and 273 replies, 50%, and `irvingoil.com` 61%. A dealership does not do business with a party it never answers. A person then approved each one. That is migration 054.

What is still worth a look, and it is a much smaller point. Some domains are still listed as customers that probably are not. Visible on the first screen of the list today:

- **linkedin.com**, 25,192 contacts, marked steady
- **peterbilt.onmicrosoft.com**, 20,043 contacts, which is the dealership itself
- **hireology.com**, a recruitment tool, and **leadforensics.com**, a marketing tool
- **amazon.ca** and **bmo.com**, a shop and a bank

These pass the reply test because somebody in the building genuinely replies to them.

The rule is sound; the list of 25 simply needs extending. That is an afternoon of somebody reading the top of the list, not a redesign.

Why some calls have no recording

This one is not our side. Of 25,154 Peterbilt recordings, **25,141 are held and playable.** Eleven are still being fetched and two were purged.

Where a recording is genuinely gone, the phone platform answers CMN-102, resource not found. The recording existed, the platform no longer has it.

The bigger reason a call has no audio is the phone company’s retention rule. TELUS keeps a call recording for **90 days, rolling**, then deletes it. This project fetches recordings and keeps them permanently, so the archive only begins where the fetching began: call records reach back to June 2025, but audio starts **7 May 2026**. Everything older was already gone.

This is not history, it is ongoing. Any recording not collected within 90 days is lost permanently, and no later run can recover it. That makes the recording job the one piece of this system where a quiet failure has a deadline attached: a week of downtime is a week of conversations that no longer exist anywhere.

Faults found in the existing system

WHAT IS WRONG	SIZE OF IT	WHY IT MATTERS
An unused column, <code>call_insight.repeat_caller</code> , is never filled in	0 of 24,560	The repeat-caller screen itself works correctly , reading a different view. This column is leftover weight beside it
Call dates cluster on the day they were imported	87%	Reading them as “when the call happened” is wrong
The “wait time” measure is always zero	99.9995%	A dead number on a live screen
Hour-of-day is stored in UTC	3 hours off	Every “when do customers contact us” answer is shifted

One thing we suspected and confirmed is not a fault. Several of the shared report views hold both dealerships' rows in the same table, and an early draft of this page called that a data-separation problem. It is not. Every query the application makes filters by dealership before showing anything (`WHERE tenant_key = $1`, checked in the code). The mixed rows were an artefact of how we exported the data for this analysis, not of the product.

Two alarms were raised by the analyses and rejected on checking. One reported that 20 mailboxes were silently missing 99.5% of their mail. On verification this is a known quirk of how coverage is measured, the count includes mail from before 2020, which was never in scope, so a dormant mailbox reads "nothing imported" for ever. It was investigated and explained on 20 August.

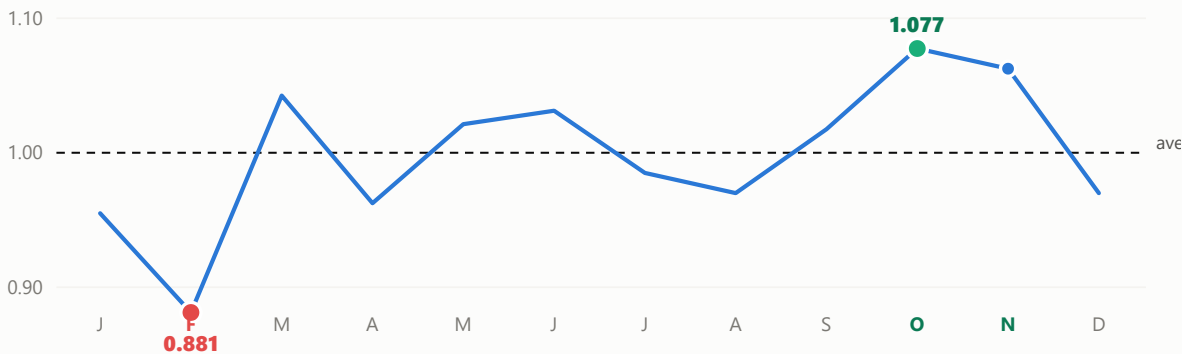
The second flagged an unused review step as a fault. It is a **deliberate removal**, decided on 8 August.

Both are recorded here because a finding that was checked and dropped says as much about the method as one that survived.

6. The business has a real annual rhythm

When the year is busy, and when it is not

Email volume by month, averaged across five complete years (2021–2025). The line at 1.00 is an average month.



⦿ COUNTED: ALL OF IT

February is the quietest month, October and November the busiest, the same shape in every one of the five years, and it holds even after allowing for months having different numbers of days. Accounts that are *growing* put more of their year into the autumn peak than fading ones do, which suggests autumn is when relationships are won or lost.

7. What we would like to ask permission for

1. May we read the complete records, rather than 0.26% of them?

Everything in Sections 1 to 3 was found by reading **10,320 emails out of 2,278,720**. The counting covered everything; the reading barely started. The table below is what changes.

WHAT WE HAVE TODAY	WHAT THE FULL CORPUS GIVES
Departure warnings on 4,611 accounts that pass a strict test	All 21,453 real accounts, with a threshold per customer type instead of one blunt rule
A complaint vocabulary from 3,627 calls	All 25,100, and a measurable <i>resolution</i> rate, not just a described one

WHAT WE HAVE TODAY

WHAT THE FULL CORPUS GIVES

"Derate" and a handful of other urgency words

The whole trade vocabulary, ranked, and the ones we have not thought of

Repeat-failure complaints customers volunteer

Tied to a specific part and a specific truck

2. Could somebody extend the automated-sender list?

Section 5. The rule already works and 25 senders are classified. A handful more are visible on the first screen of the customer list today, LinkedIn and the dealership's own domain among them. **Somebody who knows the business needs to read the top of the list and say which are customers.** It is an afternoon, not a project.

3. Could the CDK access be chased?

Section 4 shows, measured, that phone and email **cannot be connected without it.** Until then every insight here is about contact, and none is about money.

And three things can be built right now, with no new access at all:

- The **callback-owed queue** (Section 1)
- The **quiet-account feed** (Section 2), this is the "client feed" from the original brief
- The **urgency vocabulary** (Section 3)

Words used on this page

Full corpus

Every record, counted. Nothing estimated.

Sample

A slice whose text was actually read. Chosen to be representative, spread evenly across all 80 months, with extra weight on unusual

accounts, and a plain random group kept as a comparison so we can tell normal from exceptional.

Derate	A truck limiting its own power after detecting a fault. Still moving, barely. Urgent to the driver, invisible to a generic system.
Gone / fading / quiet / steady / growing	Labels the system already applies to an account based on whether contact is rising or falling against the same period a year earlier.
Voicemail never returned	No outgoing call to that number is recorded after the message was left. It is possible somebody rang from a mobile and it was not logged, that limit is stated rather than hidden.
CDK	The dealer management system that holds invoices, what customers actually bought and paid.
Seasonal index	A month compared with an average month of the same year. 1.00 is average; 0.88 is 12% below.

Every figure traces to a file. The eight raw analyses, the extraction record, the population-versus-sample counts and the verification queries with their output are all kept alongside this document. Measured 24 August 2026 against the live database, read-only. Nothing was written to it.